

Numerical Experiments for Public Good Provision under Locally Private Signals

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Abstract

This document describes the R code used to generate the numerical results for the paper *Public Good Provision under Locally Private Signals*. It records the authoritative scripts, software requirements, replication instructions, parameter values, output files, and the interpretation of each numerical experiment. Unless the Fredholm transfer-recovery experiment is explicitly discussed, all reported revenue quantities are reduced-form revenues.

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1 Authoritative Scripts

The final versions of the experiment files are:

```
exp1(3).R
exp2(3).R
exp3(4).R
exp4(4).R
exp5(4).R
exp6(4).R
exp7(4).R
exp8(4).R
exp9(2).R
```

Earlier files with the same experiment number are preliminary or intermediate versions and should not be used to reproduce the reported results.

For a cleaner replication directory, the final files may be renamed as:

```
exp1.R
exp2.R
exp3.R
exp4.R
exp5.R
exp6.R
exp7.R
exp8.R
exp9.R
```

The code does not depend on the filenames.

2 Recommended Directory Structure

A convenient repository structure is:

```
project/
|-- README.tex
|-- code/
|   |-- exp1.R
|   |-- exp2.R
|   |-- exp3.R
|   |-- exp4.R
|   |-- exp5.R
|   |-- exp6.R
|   |-- exp7.R
|   |-- exp8.R
|   |-- exp9.R
|-- results/
|   |-- exp1/
|       |-- figures/
```

```
| | '-- tables/
| |-- exp2/
| |-- ...
| '-- exp9/
|-- figures/
|-- main10.tex
|-- supplement.tex
'-- reference.bib
```

Each R script contains a machine-specific output-directory declaration. Replace that path before running the code. For example, replace:

```
out_dir <- paste0(
  "/Users/behroozmoosavi/Desktop/privacy/codes/",
  "R_codes/experiment/exp1"
)
```

with a portable declaration such as:

```
out_dir <- file.path(
  getwd(),
  "results",
  "exp1"
)
```

Each script creates its **figures** and **tables** subdirectories when necessary.

3 Software Requirements

The experiments are written in R. A recent installation is recommended:

```
R version 4.3 or later
```

The scripts use combinations of the following packages:

```
ggplot2
dplyr
tidyr
tibble
scales
```

They may be installed manually with:

```
install.packages(
  c(
    "ggplot2",
    "dplyr",
    "tidyr",
    "tibble",
```

```
    "scales"  
  )  
)
```

The figures are saved using the Cairo PDF device. If Cairo support is unavailable, replace `device = cairo_pdf` by a PDF device supported by the local R installation.

4 Running the Experiments

Each experiment is self-contained and can be run independently.

From an R session:

```
source("code/exp1.R")  
source("code/exp2.R")  
source("code/exp3.R")  
source("code/exp4.R")  
source("code/exp5.R")  
source("code/exp6.R")  
source("code/exp7.R")  
source("code/exp8.R")  
source("code/exp9.R")
```

From a terminal:

```
Rscript code/exp1.R  
Rscript code/exp2.R  
Rscript code/exp3.R  
Rscript code/exp4.R  
Rscript code/exp5.R  
Rscript code/exp6.R  
Rscript code/exp7.R  
Rscript code/exp8.R  
Rscript code/exp9.R
```

Most scripts contain:

```
FAST <- FALSE  
SHOW_TITLES <- FALSE
```

Use `FAST <- TRUE` only for debugging. All reported paper and supplement results should be generated using:

```
FAST <- FALSE  
SHOW_TITLES <- FALSE
```

5 Random Seeds

Experiment	Seed
1	Deterministic
2	20260609
3	20260609
4	20260609
5	20260609
6	20260609
7	20260610
8	20260610
9	20260612

Table 1: Random seeds used in the final experiment scripts.

The seeds should not be changed when reproducing the reported results.

6 Common Numerical Procedures

6.1 Posterior Quadrature

For a prior density f , channel density $k(y | x)$, and integrable function h , the posterior expectation is

$$\mathbb{E}[h(X) | Y = y] = \frac{\int_{\mathcal{X}} h(x) k(y | x) f(x) dx}{\int_{\mathcal{X}} k(y | x) f(x) dx}. \quad (1)$$

The scripts evaluate these integrals by deterministic trapezoidal quadrature on dense type grids.

To avoid numerical underflow, likelihoods are evaluated on the log scale. For each y , the largest log likelihood over the type grid is subtracted before exponentiation. The normalization cancels from the posterior ratio.

6.2 Posterior Interpolation

Posterior means and posterior virtual values are first calculated on dense channel-specific signal grids. Their values at simulated signal realizations are then obtained by interpolation.

The scripts report overflow diagnostics whenever a simulated signal falls outside the stored interpolation grid.

6.3 Laplace Tail Pooling

For a bounded type space

$$[\underline{x}, \bar{x}]$$

and additive Laplace noise, the posterior distribution is exactly constant on each global signal tail:

$$y \leq \underline{x}, \quad y \geq \bar{x}.$$

The numerical interpolation preserves these exact pooled-tail values.

6.4 Common Random Numbers

When channels are compared, the scripts use the same type draws and, whenever possible, the same underlying uniforms.

For $U \sim \text{Unif}(0, 1)$, standardized Laplace noise is generated by

$$L(U) = \begin{cases} \log(2U), & U < 1/2, \\ -\log(2(1 - U)), & U \geq 1/2, \end{cases}$$

while standardized logistic noise is generated by

$$G(U) = \log\left(\frac{U}{1 - U}\right).$$

This produces paired channel comparisons and reduces Monte Carlo variance.

6.5 Monte Carlo Standard Errors

For

$$\hat{\theta} = \frac{1}{B} \sum_{b=1}^B H_b,$$

the reported Monte Carlo standard error is

$$\widehat{\text{se}}(\hat{\theta}) = \sqrt{\frac{\widehat{\text{Var}}(H_b)}{B}}.$$

For paired channel differences, the variance is calculated from the observation-level paired differences.

6.6 Meaning of Revenue

Unless Experiment 5 is being discussed, all reported revenue quantities are reduced-form virtual-surplus revenues. The simulations do not generally prove that the corresponding envelope transfer belongs to the range of the channel's conditional-expectation operator.

7 Experiment 1: Posterior-Score Curves

Script: `exp1(3).R`

The prior is

$$X \sim \text{Unif}[-1, 1], \quad J(x) = 2x - 1.$$

The channels are:

$$Y^G = X + \eta^G, \quad \eta^G \sim \mathcal{N}(0, 0.4^2),$$

$$Y^L = X + \eta^L, \quad \eta^L \sim \text{Lap}(0, 0.5),$$

and

$$Y^{\text{Log}} = X + \eta^{\text{Log}}, \quad \eta^{\text{Log}} \sim \text{Logistic}(0, 0.5).$$

For

$$\lambda \in \{0, 0.3, 0.7, 1.5\},$$

the script evaluates

$$S(y, \lambda) = \hat{x}(y) + \lambda \hat{J}(y).$$

The experiment is deterministic and does not use Monte Carlo sampling.

Main conclusions:

- (i) Gaussian scores are strictly increasing at finite signals.
- (ii) Logistic scores are strictly increasing at finite signals.
- (iii) Laplace scores are weakly increasing.
- (iv) Laplace scores are exactly constant in the global signal tails.
- (v) The flat regions represent posterior pooling, not nonmonotonicity.

Outputs:

```
figures/fig_exp1_posterior_score_panel.pdf  
  
tables/exp1_posterior_score_curves.csv  
tables/exp1_zero_crossings.csv  
tables/exp1_sanity_report.csv
```

8 Experiment 2: Knife-Edge Provision Transition

Script: `exp2(3).R`

The prior is

$$X \sim \text{Unif}[-0.5, 1.5].$$

Hence

$$\mathbb{E}[X] = 0.5, \quad \mathbb{E}[J(X)] = -0.5,$$

and

$$\mu_S(\lambda) = 0.5 - 0.5\lambda.$$

The critical multiplier is

$$\lambda_{\text{crit}} = 1.$$

This must not be denoted by λ^* , since λ^* is reserved for the optimal Lagrange multiplier in the theoretical problem.

The channels are Gaussian, Laplace, and logistic, with the scales used in Experiment 1.

Population sizes:

$$n \in \{25, 50, 100, 200, 500\}.$$

Multiplier grid:

$$\lambda = 0, 0.05, 0.10, \dots, 2.$$

Monte Carlo replications:

$$B = 10,000.$$

The allocation rule is

$$q_\lambda(Y) = \mathbf{1} \left\{ \sum_{i=1}^n S(Y_i, \lambda) \geq 0 \right\}.$$

The local standardized coordinate is

$$u_n = -\frac{\sqrt{n} \mu_S(\lambda)}{\sigma_S(\lambda_{\text{crit}})},$$

and the local Gaussian approximation is

$$\mathbb{P}(q_\lambda = 1) \approx \Phi(-u_n).$$

Outputs:

```
figures/fig_exp2_knife_edge_transition.pdf
figures/fig_exp2_knife_edge_collapse.pdf

tables/exp2_knife_edge.csv
tables/exp2_dispersion.csv
tables/exp2_transition_sanity.csv
```

9 Experiment 3: Square-Root Maximum Revenue

Script: `exp3(4).R`

The prior is

$$X \sim \text{Unif}[0, 1],$$

so

$$J(x) = 2x - 1, \quad \mathbb{E}[J(X)] = 0.$$

The channels are Gaussian, Laplace, and logistic.

Population sizes:

$$n \in \{25, 50, 100, 200, 500, 1000, 2000\}.$$

Numerical settings:

One-agent moment draws	200,000
Revenue replications	10,000
Batch size	500

The simulated object is

$$R_n^{*,\text{red}} = \mathbb{E} \left[\left(\sum_{i=1}^n \hat{J}(Y_i) \right)_+ \right].$$

The first diagnostic compares

$$\frac{R_n^{*,\text{red}}}{\sqrt{n}}$$

with

$$\frac{\sigma_J}{\sqrt{2\pi}}.$$

The second diagnostic plots

$$\left(R_n^{*,\text{red}} \right)^2$$

against n . Under the square-root law, the leading slope is

$$\frac{\sigma_J^2}{2\pi}.$$

Outputs:

```
figures/fig_exp3_sqrt_n_revenue.pdf
figures/fig_exp3_rate_squared.pdf

tables/exp3_one_agent_moments.csv
tables/exp3_knife_edge_revenue.csv
tables/exp3_sanity_report.csv
```

The correct second figure filename is `fig_exp3_rate_squared.pdf`.

10 Experiment 4: Large-Deviation Rate

Script: `exp4(4).R`

The prior and multiplier are

$$X \sim \text{Unif}[-0.5, 1.5], \quad \lambda = 1.5.$$

The one-agent score mean is

$$0.5 + 1.5(-0.5) = -0.25.$$

Only the Gaussian channel is used.

Noise grid:

$$\sigma \in \{0.20, 0.30, 0.40, 0.50, 0.60, 0.80, 1.00, 1.20\}.$$

Population sizes:

$$n \in \{20, 40, 60, 80, 120, 160\}.$$

Numerical settings:

One-agent rate draws	600,000
Provision replications	30,000
Batch size	1,000

For each σ , the estimated rate is

$$I_\sigma(0) = \sup_{t \geq 0} \left\{ -\log \mathbb{E}[e^{tS_\sigma}] \right\}.$$

Outputs:

```
figures/fig_exp4_implementation_vs_sigma.pdf
figures/fig_exp4_logaccept_vs_n.pdf
figures/fig_exp4_finite_n_rate.pdf
figures/fig_exp4_rate_vs_sigma2.pdf

tables/exp4_one_agent_rate.csv
tables/exp4_implementation_probabilities.csv
tables/exp4_finite_n_rate.csv
tables/exp4_sanity_report.csv
```

The fitted relationship between the estimated rate and σ^2 is descriptive over the selected range. It is not a universal asymptotic theorem.

11 Experiment 5: Fredholm Transfer Inversion

Script: `exp5(4).R`

The prior and mechanism settings are

$$X \sim \text{Unif}[0, 1], \quad n = 50, \quad \lambda = 1.$$

Gaussian noise grid:

$$\sigma \in \{0.10, 0.20, 0.40, 0.70, 1.00\}.$$

Discretization:

Type-grid points	360
Signal-grid points	240
Quadrature points	5,001

The discretized matrix is deliberately tall.

The regularized estimator solves approximately

$$\hat{g}_\rho = \arg \min_g \left\{ \|K_h g - T_h\|_2^2 + \rho \|D_2 g\|_2^2 + \zeta \|g\|_2^2 \right\}.$$

Regularization settings:

$$\rho_{\text{main}} = 10^{-4}, \quad \rho \in \{10^{-5}, 10^{-4}, 10^{-3}\}.$$

The relative target perturbation is one percent. The primary stability calculation uses 200 perturbations, and the regularization sweep uses 80 perturbations per value of ρ .

Outputs:

```
figures/fig_exp5_conditioning.pdf
figures/fig_exp5_stability_amplification.pdf
figures/fig_exp5_fredholm_fit.pdf
figures/fig_exp5_singular_values.pdf

tables/exp5_envelope_target_vs_implemented.csv
tables/exp5_singular_values.csv
tables/exp5_summary.csv
tables/exp5_rho_sweep.csv
```

A full-column-rank discretized matrix is not proof of completeness of the continuous operator. A small finite-grid residual does not establish exact continuous range membership.

12 Experiment 6: Hierarchical Unknown Endpoint

Script: `exp6(4).R`

The lower support point, true endpoint, and hyperparameter support are

$$x_0 = 0.3, \quad \theta_0 = 1.5, \quad \theta \in [1, 2].$$

The channel is

$$Y = X + \eta, \quad \eta \sim \mathcal{N}(0, 0.45^2),$$

and

$$\lambda = 0.5.$$

Conditional types are generated as

$$X = 0.3 + (\theta - 0.3)Z.$$

Two specifications are used:

$$Z \sim \text{Beta}(1, 1),$$

labelled **Uniform**, and

$$Z \sim \text{Beta}(2, 2),$$

labelled **Beta(2,2)**.

Population sizes:

$$n \in \{25, 50, 100, 200, 400, 800\}.$$

Monte Carlo replications:

$$B = 4,000.$$

The posterior over θ uses an 81-point grid.

Final suffixes:

uniform, **beta.**

Outputs:

```
figures/fig_exp6_revenue_levels_uniform.pdf
figures/fig_exp6_revenue_relative_gap_uniform.pdf
figures/fig_exp6_theta_posterior_concentration_uniform.pdf

figures/fig_exp6_revenue_levels_beta.pdf
figures/fig_exp6_revenue_relative_gap_beta.pdf
figures/fig_exp6_theta_posterior_concentration_beta.pdf
```

Because the support $[0.3, \theta]$ depends on θ , the experiment lies outside the common-support hierarchical theorem. Posterior-concentration slopes are descriptive.

13 Experiment 7: Approximate-LDP Comparison

Script: `exp7(4).R`

The active channels are:

Gaussian, Laplace, logistic.

The operative privacy parameter is

$$\delta = 10^{-5}.$$

The executable code setting

```
privacy_delta <- 1e-5
```

is authoritative. Any older comment mentioning 10^{-6} is stale.

Privacy grid:

$$\epsilon \in \{0.25, 0.50, 0.75, 1, 1.5, 2, 3, 4, 6\}.$$

Population and Monte Carlo settings:

$$n = 100, \quad B = 20,000.$$

Welfare design:

$$X \sim \text{Unif}[-1, 1], \quad \lambda = 0.$$

Revenue design:

$$X \sim \text{Unif}[-0.5, 1.5], \quad \lambda = 1.$$

At $\epsilon = 1$, the multiplier grid is

$$\lambda \in \{0, 0.3, 0.6, 1, 1.3, 1.6, 2\}.$$

Outputs:

```
figures/fig_exp7_efficiency_vs_eps.pdf
figures/fig_exp7_revenue_vs_eps.pdf
figures/fig_exp7_objective_vs_lambda.pdf

tables/exp7_privacy_calibration.csv
tables/exp7_welfare_vs_eps.csv
tables/exp7_revenue_vs_eps.csv
tables/exp7_vs_lambda.csv
tables/exp7_sanity_report.csv
```

The revenue measure is signed reduced-form revenue under the specified score rule. It is not truncated at zero.

14 Experiment 8: Welfare–Revenue Frontiers

Script: `exp8(4).R`

Only Laplace and logistic channels are used.

The prior is

$$X = 2U, \quad U \sim \text{Beta}(2, 2),$$

so $X \in [0, 2]$.

The equal-scale calibration is

$$b_L = \beta = \frac{2}{\epsilon}.$$

Privacy grid:

$$\epsilon \in \{1, 2, 4, 6, 8, 10\}.$$

Population and Monte Carlo settings:

$$n = 100, \quad B = 20,000.$$

For each simulated observation, define

$$A = \sum_i \hat{x}(Y_i), \quad B_J = \sum_i \hat{J}(Y_i).$$

The rule builds when

$$A + \lambda B_J \geq 0.$$

If $B_J < 0$, the observation switches at

$$\lambda^{\text{switch}} = -\frac{A}{B_J}.$$

The script sorts the observation-specific switching thresholds and constructs the finite-sample multiplier frontier directly. It does not use a coarse multiplier grid or an ex-post cumulative-maximum correction.

Outputs:

```
figures/fig_exp8_welfare_vs_revenue_target.pdf
figures/fig_exp8_welfare_vs_revenue_target_byeps.pdf
figures/fig_exp8_Rmax_vs_eps.pdf

tables/exp8_welfare_vs_revenue_target.csv
tables/exp8_Rmax_by_channel_eps.csv
tables/exp8_empirical_frontier.csv
tables/exp8_sanity_report.csv
```

At equal scale, Laplace Blackwell-dominates logistic, so the displayed ordering is consistent with the formal theory.

15 Experiment 9: Three Privacy Standards

Script: `exp9(2).R`

The continuous prior is

$$X \sim \text{Unif}[-1, 1].$$

Population and Monte Carlo settings:

$$n = 100, \quad B = 10,000.$$

Approximate-LDP parameter:

$$\delta = 10^{-5}.$$

Privacy-index grid:

$$\epsilon \in \{0.5, 0.75, 1, 1.5, 2, 3\}.$$

Endpoint GDP grid:

$$\mu \in \{0.5, 1, 2\}.$$

15.1 Axis 1: Common Pure- ϵ Scale

$$b_L = \beta = \frac{2}{\epsilon}.$$

Only Laplace and logistic channels are compared. The theory-oriented welfare gap is

$$W_{\text{Lap}} - W_{\text{Log}}.$$

15.2 Axis 2: Common (ϵ, δ) -LDP Boundary

Gaussian, Laplace, and logistic channels are calibrated to the same hockey-stick privacy boundary with $\delta = 10^{-5}$. No universal Blackwell ranking is asserted.

15.3 Axis 3: Common Tight μ -GDP

The scales are

$$\sigma^*(\mu) = \frac{2}{\mu},$$

$$b_L^*(\mu) = \frac{2}{-2 \log(2\Phi(-\mu/2))},$$

and

$$\beta^*(\mu) = \frac{2}{2 \log(\Phi(\mu/2)/\Phi(-\mu/2))}.$$

For the endpoint comparison, the theory-oriented welfare gap is

$$W_{\text{Log}} - W_{\text{Lap}}.$$

The endpoint state distribution is

$$X = \begin{cases} 1, & \text{with probability } \pi_H, \\ -1, & \text{with probability } 1 - \pi_H. \end{cases}$$

Outputs:


```

figures/fig_exp9_tradeoff_common_mu.pdf
figures/fig_exp9_tradeoff_gap_common_mu.pdf
figures/fig_exp9_endpoint_welfare.pdf

figures/fig_exp9_welfare_axis1_equal_epsilon.pdf
figures/fig_exp9_gap_axis1_equal_epsilon.pdf

figures/fig_exp9_welfare_axis2_eps_delta.pdf
figures/fig_exp9_gap_axis2_eps_delta.pdf

figures/fig_exp9_welfare_axis3_common_mu.pdf
figures/fig_exp9_gap_axis3_common_mu.pdf

```

Principal tables:

```

tables/exp9_tradeoff_common_mu.csv
tables/exp9_tradeoff_dominance_summary.csv
tables/exp9_endpoint_welfare.csv
tables/exp9_endpoint_welfare_dominance_summary.csv
tables/exp9_gdp_calibration_checks.csv
tables/exp9_welfare_three_axes.csv
tables/exp9_laplace_minus_logistic_paired_gap.csv
tables/exp9_overflow_diagnostics.csv

```

The endpoint trade-off and endpoint welfare rankings are theorem-backed. The continuous-type common- μ gaps are numerical and specification dependent.

16 Main-Paper Figure Mapping

Main-paper object	Generated filename
Posterior-score panel	fig_exp1_posterior_score_panel.pdf
Knife-edge collapse	fig_exp2_knife_edge_collapse.pdf
Square-root revenue	fig_exp3_sqrtn_revenue.pdf
Equal-scale welfare gap	fig_exp9_gap_axis1_equal_epsilon.pdf
Endpoint welfare comparison	fig_exp9_endpoint_welfare.pdf

Table 2: Mapping from the numerical discussion in the main paper to the generated figure files.

17 Interpretation Cautions

17.1 Reduced Form versus Exact Implementation

Experiments 2–4 and 7–9 calculate reduced-form allocations, welfare, or virtual-surplus revenue. They do not establish that the associated envelope transfers belong to the range of the privacy

channel’s conditional-expectation operator.

Experiment 5 studies a discretized and regularized approximation to that inverse problem.

17.2 Hierarchical Experiment

Experiment 6 has a parameter-dependent support and therefore lies outside the formal common-support hierarchical theorem.

17.3 Channel Rankings

The following rankings are supported by formal theory:

- (i) At equal raw scale, Laplace Blackwell-dominates logistic over the full additive location experiment.
- (ii) At common tight μ -GDP, logistic Blackwell-dominates Laplace on the binary endpoint experiment.

The continuous-type common- μ comparison is numerical and does not establish a complete Blackwell order.

17.4 Large-Deviation Regression

The estimated relationship between the Cramér rate and Gaussian noise variance in Experiment 4 is descriptive over the selected parameter range.

18 Recommended Replication Workflow

1. Place the nine authoritative scripts in a clean `code` directory.
2. Rename the scripts to `exp1.R`, `...`, `exp9.R` if desired.
3. Replace every machine-specific `out_dir`.
4. Set `FAST <- FALSE`.
5. Set `SHOW_TITLES <- FALSE`.
6. Run the scripts in numerical order.
7. Review the sanity report produced by every script.
8. Confirm that interpolation-overflow rates are negligible.
9. Confirm the tower identities

$$\mathbb{E}[\hat{x}(Y)] \approx \mathbb{E}[X], \quad \mathbb{E}[\hat{J}(Y)] \approx \underline{x}.$$

10. Confirm posterior-score monotonicity.
11. Confirm the crossing near $\lambda_{\text{crit}} = 1$ in Experiment 2.
12. Confirm convergence toward $\sigma_J/\sqrt{2\pi}$ in Experiment 3.
13. Compare finite- n and one-agent rate estimates in Experiment 4.
14. Review singular-value decay, residuals, and perturbation amplification in Experiment 5.
15. Treat the hierarchical contraction slopes in Experiment 6 as descriptive.
16. Verify approximate-LDP calibration residuals in Experiment 7.
17. Verify the equal-scale Laplace inequalities in Experiment 8.
18. Verify endpoint trade-off and welfare dominance in Experiment 9.
19. Copy the final PDF figures into the manuscript **figures** directory.
20. Compile the main paper and Online Supplement.

19 Suggested Citation

The accompanying paper may be cited as:

```
@article{moosavi_awan_public_good_privacy,
  title = {Public Good Provision under Locally Private Signals},
  author = {Moosavi, Behrooz and Awan, Jordan},
  year = {2026},
  note = {Working paper}
}
```

20 Contact

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